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The Impact Of Mining Sector Investment On National And Regional Economic Performance

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M.Rizal Taufikurahman[1], Ahmad Heri Firdaus[2], Tauhid Ahmad[3], Dyah Ayu Febriani[4], Agung Satria Permana[5]

[1] *Head Centre of Macroeconomics and Finance INDEF*

[2] *Researcher at Centre of Industry, Trade and Investment INDEF*

[3] *Executive Director of INDEF*

[4] *Research Assistant of INDEF*

[5] *Research Assistant of INDEF*

EXECUTIVE SUMMARY

Indonesia is one of the countries with high potential reserves of mining goods, such as nickel, coal, oil and gas, tin ore, copper, gold, and so on, due to mining activities. The potential for mining results is enormous, so the mining sector contributes highly to state revenues while still referring to sustainable principles in using natural resources for the greatest prosperity of the people and achieving Sustainable Development Goals (SDGs).

Currently, Indonesia is increasing and realizing various investments in the mining sector, especially nickel, in many regions. However, the impact of investment in the mining sector, especially nickel, on the economy, both nationally and regionally, has yet to be discovered. Therefore, this policy brief will focus on mapping the performance of investment in the nickel mining sector nationally and regionally in the provinces of Central Sulawesi, South Sulawesi, Southeast Sulawesi, and North Maluku and showing the results of calculating the impact of investment in the nickel mining sector on national, regional, and sectoral economic performance.

Using a mixed approach study design approach with general balance economic modeling or CGE model through GEMPACK software and descriptive analysis to map the development and performance of mining sector investment by region/region, socio-economic conditions, poverty, investment realization, contribution of the mining sector, especially nickel mining, as well as the ESG performance of nickel mining companies, several results found, namely:

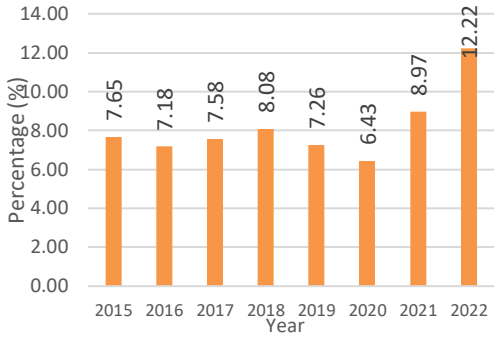
1. Every 1 percent increase in nickel mining investment and operations in Southeast Sulawesi will result in an additional 0.98 percent in national economic value (real GDP), followed by South Sulawesi at 0.92 percent, North Maluku at 0.72 percent, and Central Sulawesi at 0.92 percent 0.66 percent.
2. Every 1 percent increase in nickel ore production in South Sulawesi will result in an additional 0.08 percent in Gross Regional Domestic Product (GRDP). On the other hand, every 1 percent increase in nickel ore production input capacity in Southeast Sulawesi and North Maluku will contribute to the rise in GRDP of 0.05 percent. Meanwhile, every 1 percent increase in nickel ore input capacity in Central Sulawesi will produce an additional GRDP of 0.02 percent.
3. Every 1000-ton increase in nickel ore input capacity will increase the workforce by 47 people in South Sulawesi, 29 people in Southeast Sulawesi, 15 people in Central Sulawesi, and 7 people in North Maluku.

BACKGROUND

Investment is one of the economic indicators that improves national economic performance. Through high investment realization, economic growth will increase. In fact, investment is a necessity that cannot be rejected in development activities, especially economic development. Indonesia's investment trend is rising; even mining investment ranked first in investment realization in the fourth quarter of 2022, amounting to IDR 39.8 trillion (BKPM, 2022). The significant investment in the mining sector has shown great potential to boost the regional and national economy. This sector is also a mainstay for employment, state revenues, and foreign exchange.

Apart from that, the mining sector's contribution to the national Gross Domestic Product (GDP) continues to increase even though it experienced a decline when the COVID-19 pandemic first hit. The GDP value of the mining sector has grown in contribution from 6.43 percent in 2020 to 8.97 percent in 2021. A massive increase in contribution occurred in 2022 to 12.22 percent. This increase shows that the proportion of the mining sector to the national GDP is increasing with a very high increase in value added from this sector.

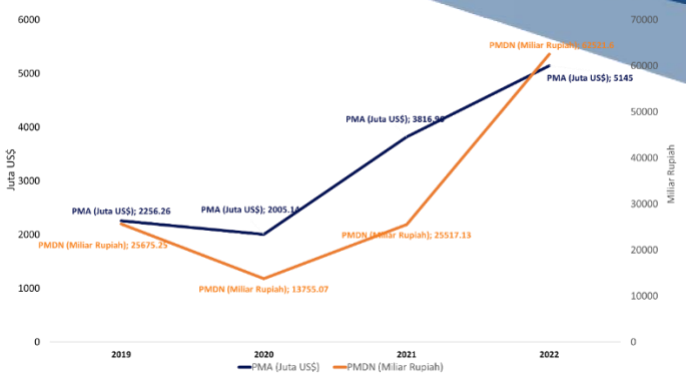
Figure 1. Growth of the Proportion of the Mining Sector to National GDP in 2015 – 2022



Source: BPS (2023)

In direct proportion to the development of the ratio of the mining sector to national GDP, the growth of the investment value of the mining sector has also increased after 2020. The investment value of the mining sector has increased, especially those originating from Domestic Investment (PMDN), which shows that the industry Mining has a reasonably high investment attraction for domestic investors aiming for export-oriented downstream extraction and production as well as meeting domestic industrial needs. The solid growth of the mining sector is also due to the government's efforts to encourage the downstream of the primary industry with high-added value and strengthen the productivity of the upstream side of the industry. Downstreaming will promote the creation of added value within the country and support the improvement of the national economy.

Figure 2. Growth of Mining Sector Investment Value Achievements 2019 – 2022 Perkembangan Capaian Nilai Investasi Sektor Pertambangan Tahun 2019 - 2022



Source: BPS, 2015-2021

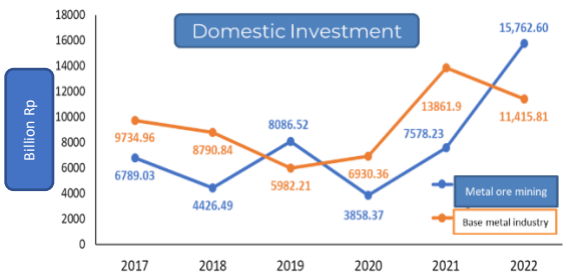
Indonesia is rich in natural resources in the mining sector, including nickel. Indonesia's nickel production in 2021 is estimated to reach 1.00 million MT in 2021 or contribute 30.40 percent of total world production, making Indonesia the world's leading producer of nickel (Mine Worldwide in Statista, 2022). In addition, based on the US Geological Survey (USGS), Indonesia has the world's largest nickel reserves, with reserves of 21 million tons or 22 percent of world reserves. The enormous nickel resources that Indonesia has must, of course, be managed well and utilized optimally for the people's welfare by the Constitution's mandate. On the other hand, after two years of implementing the Nike ore export ban policy, Indonesia is starting to feel the sweet fruit of the downstream policy implemented.

Therefore, this policy brief intends to analyze the impact of mining activities in the form of nickel, especially the investment value, and to estimate the impact of mining sector development on the national and regional economy in the provinces of South Sulawesi, Central Sulawesi, Southeast Sulawesi and North Maluku and calculate the impact investment in the nickel mining sector on national, regional and sectoral economic performance, including the impact of investment on labor and disclosure of environmental, society and governance performance that nickel mining companies have carried out.

INVESTMENT PERFORMANCE IN THE MINING SECTOR

The realization of domestic investment (PMDN) in the metal ore mining sector in 2022 increased by IDR 8.9 trillion (132 percent), and the base metal industry also increased by IDR 1.68 trillion (17 percent) compared to 2017. Meanwhile, for foreign investment (PMA), investment realization also increased in 2022, amounting to USD 1.13 billion (54 percent). In the Base Metal Industry, there was an increase of USD 8.26 billion (320 percent) compared to 2017. The significant increase in investment in the base metal industry sector indicates that investors' focus is currently starting to shift toward the sector downstream.

Figure 3. Realization of Domestic Investment (PMDN) and Foreign Investment (PMA) in the Metal Ore Mining and Basic Metal Industry Sector



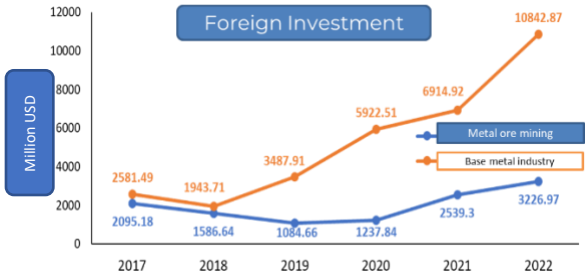
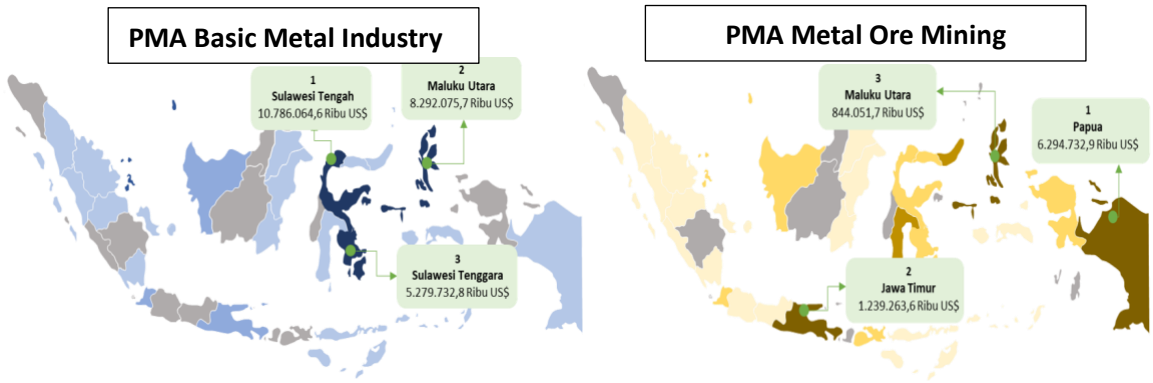


Figure 4. Realization of Domestic Investment (PMDN) and Foreign Investment (PMA) in the Metal Ore Mining and Basic Metal Industry Sector



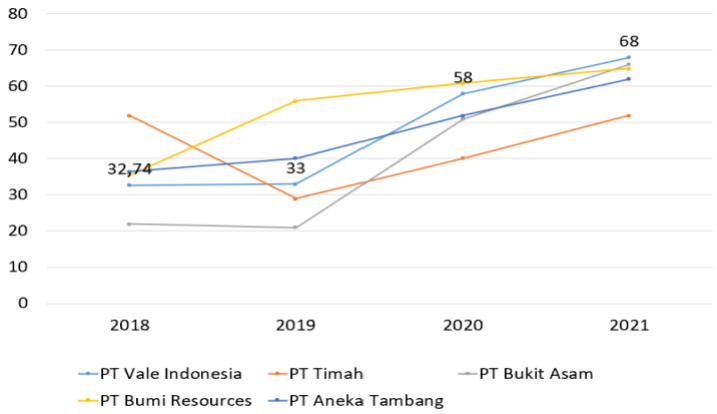
Source: BKPM (2022)

If explored more deeply, investment realization in the downstream sector (basic metal industry) dominates by the three most significant nickel producing provinces in Indonesia, namely Central Sulawesi, North Maluku, and Southeast Sulawesi, with a contribution reaching 83.35 percent, as shown in Figure 4.

MINING COMPANY ENVIRONMENT, SOCIAL, AND GOVERNANCE (ESG) DISCLOSURES

Environmental, Social, and Governance (ESG) indicators are currently widely used to measure the sustainability and ethical impact of investment results in businesses or companies. ESG can be a comprehensive framework that helps companies, especially mining companies, with a disreputable stigma regarding sustainability issues. The graph above illustrates the significant movement of each company towards ESG disclosure every year, most recently in 2021, PT. Vale is the ESG leader with a disclosure score of 68 percent.

Figure 5. ESG Disclosure According to Bumi Global Karbon Foundation in 2022



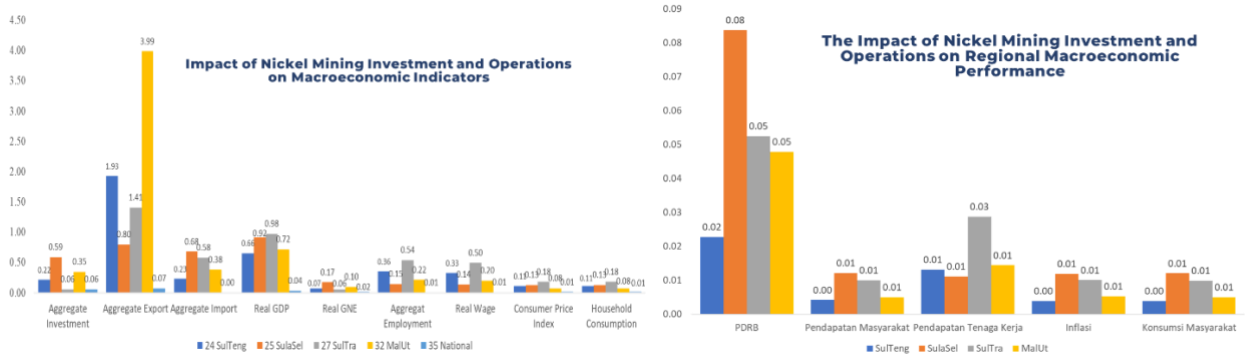
Source: BGK Foundation, 2022

Besides that, Sustainalytics also issued an ESG indicator that investors can use, called ESG risk rating (the lower the value, the better). Sustainalytics has assessed 205 world metal mining companies and 4 of them are from Indonesia, namely: 1. PT Vale Indonesia Tbk with a score of 32.1 or occupies 43rd position out of 205 companies, PT Aneka Tambang Tbk has a score of 39.9 (rank 70), PT Merdeka Copper Gold Tbk with a score of 45.8 (rank 110), and PT Timah Tbk which got a score of 56.1 or ranked 177th out of 205 companies.

THE IMPACT OF NICKEL MINING SECTOR INVESTMENTS AND OPERATIONS ON NATIONAL AND REGIONAL ECONOMIC PERFORMANCE

The impact of investment and nickel mining operations on national macroeconomic indicators can be seen from the export side; for example, every one percent increase in nickel ore input capacity in North Maluku will provide an additional 3.99 percent to the national export value. Apart from that, if we look at the contribution to the national economy through Real GDP, every 1 percent increase in nickel ore input capacity in Southeast Sulawesi will contribute the most significant impact, namely 0.98 percent, followed by South Sulawesi at 0.92 percent, North Maluku amounting to 0.72 percent, and Central Sulawesi 0.66 percent.

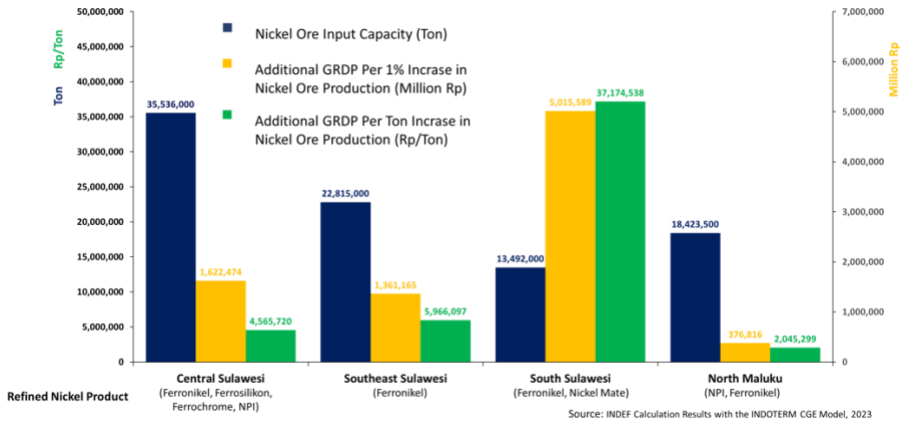
Figure 6. Impact of Nickel Mining Investment and Operations (Increase in Ore Input Capacity) on Regional Economic Indicators and National Macroeconomic Performance



Source: INDEF Calculation Results with the INDOTERM CGE Model, 2023

On the other hand, the impact of investment and nickel mining operations on regional macroeconomic performance shows that 1. Every 1 percent increase in nickel ore input capacity in South Sulawesi will result in an additional 0.08 percent GRDP. Furthermore, every 1 percent increase in nickel ore input capacity in Southeast Sulawesi and North Maluku will contribute to the rise in GRDP of 0.05 percent. Meanwhile, every 1 percent increase in nickel ore input capacity in Central Sulawesi will result in an additional GRDP of 0.02 percent.

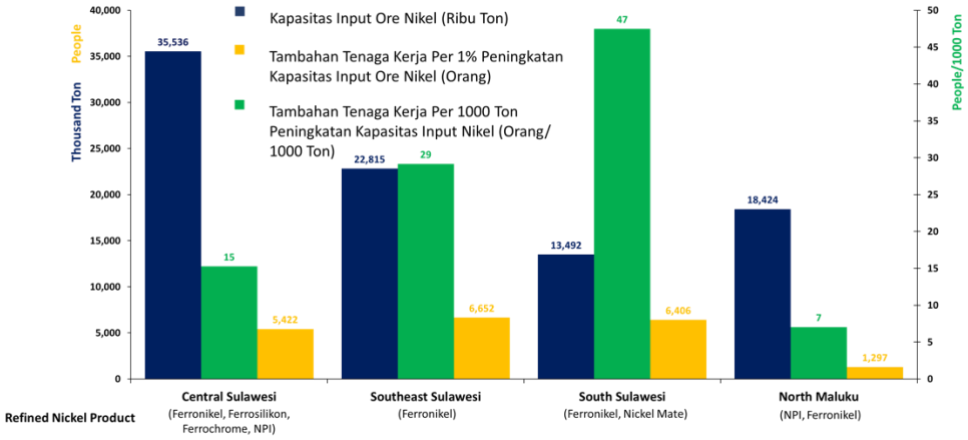
Figure 6. Impact of Nickel Mining Investment and Operations on GRDP



Source: INDEF Calculation Results with the INDOTERM CGE Model, 2023

Figure 6. Provides an overview of nickel ore input capacity (tons), additional GRDP per 1 percent increase in nickel ore input (million rupiah), and additional GRDP per ton increase in nickel ore production. South Sulawesi is the province that has the most significant impact on GRDP, with details that every 1 percent increase in nickel ore processing input will influence increasing GRDP in South Sulawesi by IDR 5,015,589 million and will have an impact on improving GRDP by IDR 37,174,538 per ton, increasing input capacity for nickel ore production.

Figure 7. Impact of Nickel Mining Investment and Operations on Labor



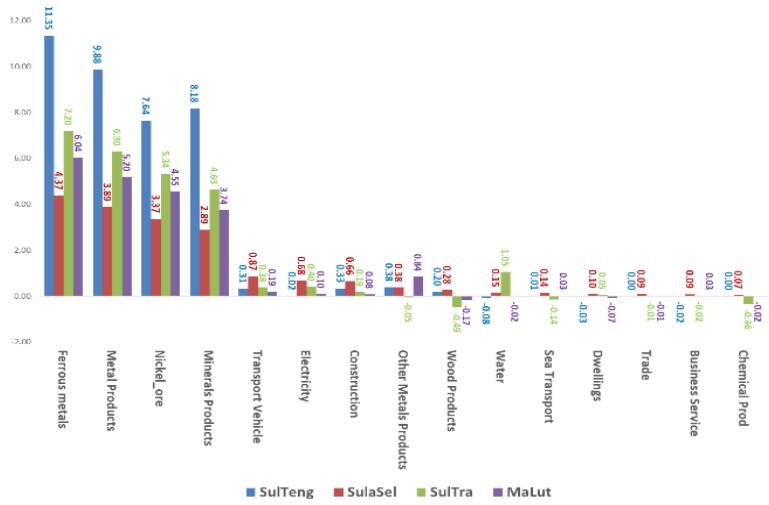
Source: INDEF Calculation Results with the INDOTERM CGE Model, 2023

Figure 7 illustrates that every 1,000-ton increase in nickel ore input capacity will increase the workforce by 47 people in South Sulawesi, 29 people in Southeast Sulawesi, 15 people in Central Sulawesi, and 7 people in North Maluku. In addition, if there is a 1 percent increase in nickel ore input capacity, it will increase the workforce of 6,652 people in Southeast Sulawesi, 6,406 people in South Sulawesi, 5,422 people in Central Sulawesi and 1,297 people in North Maluku.

THE IMPACT OF INVESTMENTS AND OPERATIONS IN THE NICKEL MINING SECTOR ON THE OUTPUT OF SECTORS

Investment in the nickel mining sector can impact the output of the nickel mining sector and its derivative products. The output of the ferrous metal sector in Central Sulawesi increased by 11.35 percent. Additionally, output in South Sulawesi Province also increased by 4.37 percent, Southeast Sulawesi increased by 7.20 percent, and North Maluku increased by 6.04 percent. Furthermore, the increase in output occurs not only in the mining sector but also in the downstream mining industry. As shown in Figure 8, the industrial sector of mineral mining products, steel, machinery, and equipment experienced an increase in output ranging from 3.89 percent to 9.88 percent. The most significant increase occurred in Central Sulawesi's Ferrous metal (steel) industry, whose output increased by 11.35 percent. Apart from that, the metal products industry in North Maluku Province also increased by 9.88 percent. Although the increase in output in South Sulawesi Province in these two sectors experienced the lowest increase compared to the other three provinces, there was the highest increase in different sectors such as in the transport vehicle sector (0.47 percent), electricity (0.68 percent), and construction (0.66 percent).

Figure 8. Impact of Nickel Mining Investment and Operations on the Output of Several Sectors



Source: INDEF Calculation Results with the INDOTERM CGE Model, 2023

IMPACT OF INVESTMENTS AND OPERATIONS OF THE NICKEL MINING SECTOR ON PRODUCT USE ACCORDING TO SOURCES

Furthermore, Figure 10 illustrates that investments and operations in the nickel mining sector will impact the composition of input use according to the source of origin of the goods. Based on the CGE results shown in Figure 10, there has been an increase in the use of domestic industrial products. The largest increase in the use of domestic products occurred in the ferrous metal sector, where it increased by 0.53 percent. At the same time, there was a decrease in the use of imported products in this industry by 0.161 percent. Futhest, the largest decline in the use of imported products occurred in non-metallic minerals products, amounting to 1.894 percent.

Figure 9. Impact of Nickel Mining Investment and Operations on Product Use According to Sources

Top 10 Sector	Domestic	Import
Ferrous metals.	0.516	-0.161
Mineral products nec.	0.488	-0.454
Non-Metallic Minerals	0.457	-1.894
Non-Ferrous Metals	0.153	-0.016
Fabricated Metal Products	0.139	-0.284
Construction.	0.061	-0.162
Chemical Rubber Products	0.025	0.027
Water	0.014	0.115
Electricity and Gas Distribution	0.011	0.088
Wood and products of wood	0.011	0.131

CHALLENGES OF OPTIMIZING INVESTMENT IN THE NICKEL MINING SECTOR IN Indonesia

Even though Indonesia is one of the countries with the largest nickel reserves in the world and is supported by the value of mining investment from both domestic and foreign sources, this sector is not free from challenges, including downstream investment, which is expensive and requires high technology and human resource skills. Rising energy costs, social and geopolitical risks, and limited infrastructure and national resources are also challenges for the mining sector; mining companies are under pressure to control costs, increase efficiency, and improve security (Deloitte, 2022; Berbner et al., 2022).

Good mining standards or practices, such as environmental protection and monitoring mining activities in ecological, social, and human rights aspects, as well as post-mining processes, need to be considered (Abdullah, M., 2015). On the other hand, providing incentives and facilities has not been optimal for the mining industry, especially nickel, which has implemented environmental, social, and governance (ESG) principles and Good Mining Practices (GMP). Another challenge is that all companies investing in nickel downstream cannot yet apply ESG principles and good mining practices.

POLICY RECOMMENDATION

1. This study produced several policy recommendations including: Investment is the key to downstream mining, where the higher the investment in downstream products, the better it is for the national and regional economy.
2. The government is expected to collaborate on research and joint ventures with private industry to strengthen the transfer of more efficient nickel downstream technology.
3. The government is expected to invest more heavily, especially in the mining industry, especially nickel mining, to increase greater added value through sustainable downstream.
4. There is a need to implement ESG and GMP principles for all mining companies in Indonesia, accompanied by providing morning incentives to all companies that comply with them and have implemented them.
5. The government is expected to implement monitoring and evaluation of the implementation of ESG and GMP principles for all mining companies to support environmental sustainability, social welfare, human rights, and justice.
6. Developing a funding scheme that supports the nickel downstream industry by ESG and GMP principles is necessary.

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